

SHIHAB BIN KADER

Chattogram, Bangladesh
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Computer Science and Engineering graduate with practical experience in data analysis, artificial intelligence, computer vision, deep learning, software development, research, and technical documentation. Skilled in translating technical problems into structured solutions, working with data, and communicating results clearly in academic and professional settings.

Academic Credentials

East Delta University, Chattogram 2020-2025
Bachelor of Science in Computer Science and Engineering
Credits Completed: 162
Relevant Coursework: Artificial Intelligence, Pattern Recognition, Digital Image Processing, Machine Learning, Big Data Analytics, Python Programming

Govt. Hazi Muhammad Mohsin College, Chattogram 2016-2018
Higher Secondary Certificate (HSC) - Science

Govt. Muslim High School, Chattogram 2014-2015
Secondary School Certificate (SSC) - Science

Undergraduate Thesis & Research

Deep Learning-Based Mango Ripeness Classification with Explainable AI | *Undergraduate Researcher, 2024-2025*

- Developed a computer-vision approach for classifying mangoes into multiple ripeness stages using a balanced dataset of approximately 6,000 images.
- Applied image preprocessing, augmentation, convolutional neural networks, transfer learning, and Grad-CAM explainability.
- Evaluated models using accuracy, precision, recall, F1-score, confusion matrices, and ROC/AUC analysis.

Publications & Research

1. Abdullah, M. S., Faiyaz, G. M. F., Forhad, O. N., Kader, S. B., & Labib, S. M. A. (2025). “Hybrid Xception-LSTM Model for Explainable Mango Ripeness Classification Using Grad-CAM.” IEEE 4th RAAICON, Dhaka, Bangladesh. Co-author.
   Co-author contribution: Supported literature review, data analysis, interpretation of results, and manuscript review/revision.
2. “Evaluation of Phytochemical Analysis and In Vitro and In Vivo Pharmacological Activity of Methanol Extract of Hyptis brevipes.” Co-author. [In Review]
   Co-author contribution: Supported literature review, data organization and analysis, interpretation of findings, and manuscript review/revision.

Selected Technical Projects

- **Sign Sense - ASL Recognition System:** Built a computer-vision application to recognize 29 American Sign Language classes using deep learning and image processing.
- **Drowsiness Detection AI:** Built a real-time computer-vision system to identify signs of driver drowsiness.
- **ModelDNA:** Applied explainable AI methods to analyze and interpret model decisions.
- **Image Caption Generator:** Built a deep-learning application for image understanding and natural-language generation.
- **Multimodal AI Application:** Combined information from multiple data modalities for prediction and analysis.
- **Fun with Flags:** Developed an Android application for interactive flag recognition and learning.

Technical Skills

- **Programming, Data & Office:** Python, C, C++, Java, R, SQL; MySQL, PostgreSQL; Excel, Power BI, Word, PowerPoint
- **AI, Computer Vision & Frameworks:** Machine Learning, Deep Learning, Computer Vision, Explainable AI, CNN, Transfer Learning, Grad-CAM; PyTorch, TensorFlow, Keras, OpenCV, Pandas, NumPy
- **Development & Application Tools:** Git, GitHub, Docker, Android Studio, Streamlit, FastAPI

Extra-Curricular Activities

- **Executive Member | Computer Club, East Delta University**
- **Community Volunteer | Mayaful, Chattogram**
  - Supported educational and social programs for children and community welfare activities.

Academic Achievements

- Dean’s List, East Delta University
- Recipient of East Delta University Academic Scholarship

Language Proficiency

IELTS Academic: Overall Band 7.5 (CEFR C1) | Listening 8.0 | Reading 8.0 | Writing 6.5 | Speaking 7.0
Languages: Bengali - Native | English - Professional Working Proficiency

Professional Strengths

Analytical problem solving | Team collaboration | Technical communication | Adaptability

References

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